# Quiz — 1.1.3 Input, Output and Storage Devices

Name: ____ Date: _ ***Mark:*** ___ / 25

*OCR H446 · Component 01 · 1.1.3*

## Section A — Multiple choice (8 marks)

**1.** Which of the following is an **output** device? A. Barcode scanner B. Microphone C. Actuator D. Capacitive touchscreen (used to type) Your answer: ☐

**2.** A **magnetic hard disk drive (HDD)** stores data by: A. Trapping electrons in cells with no moving parts B. Magnetising patterns/polarity on a spinning platter, read by a R/W head C. Using a laser to detect pits and lands D. Holding data only while powered Your answer: ☐

**3.** Which storage type uses a **laser to detect pits and lands**? A. Magnetic B. Flash / SSD C. Optical (CD/DVD/Blu-ray) D. RAM Your answer: ☐

**4.** A key advantage of **flash/SSD** storage over magnetic HDDs is that it: A. Has the lowest cost per gigabyte B. Has no moving parts, so it is faster, silent and more durable C. Has unlimited write cycles D. Has the largest possible capacity Your answer: ☐

**5.** Which statement correctly describes **RAM**? A. Non-volatile and normally read-only B. Volatile, can be read from and written to, holds running programs/data C. Holds the bootstrap/BIOS firmware permanently D. A type of secondary storage Your answer: ☐

**6.** **ROM** is typically used to store: A. The currently running programs B. Documents the user is editing C. The bootstrap loader / BIOS firmware D. Temporary clipboard data Your answer: ☐

**7.** **Virtual memory** is best described as: A. Remote storage accessed over the internet B. Cache memory inside the CPU C. The use of secondary storage as extra RAM when physical RAM is full D. A faster type of RAM Your answer: ☐

**8.** Magnetic **tape** is described as a **serial access** medium, making it most suitable for: A. Random access of individual records very quickly B. Cheap, high-capacity archive/backup C. Running an operating system D. High-speed gaming storage Your answer: ☐

## Section B — Short answer (12 marks)

**9.** State **two** differences between **RAM** and **ROM**. **[2]**

**10.** Explain why **secondary storage** is described as **non-volatile**, and give **one** example of a secondary storage device. **[2]**

**11.** A user needs portable storage to carry large video files between sites, frequently rewriting them, with fast transfer and good durability. Recommend a suitable storage type and **justify** your choice against the requirements. **[3]**

**12.** Explain how **virtual memory** works and state **one** drawback of relying on it heavily. **[2]**

**13.** Define the term **disk thrashing**. **[1]**

**14.** State **two** advantages of using **virtual (cloud) storage** instead of local storage. **[2]**

## Section C — Exam-style question (5 marks)

**15.** A busy office needs a printer to produce a high volume of black-and-white text documents quickly and cheaply each day. A separate home user occasionally prints a few colour photographs.

Recommend a suitable type of printer for **each** user and justify your choices by comparing the two printer types against the stated requirements. **(AO2/AO3) [5]**

Office: ______

Home: ______

## Answer key

**Section A (8 marks — 1 each)**

1. **C** — An actuator is an output device (it acts on the physical world); the others are inputs.
2. **B** — HDDs magnetise patterns on a spinning platter read by a read/write head.
3. **C** — Optical media use a laser to detect pits and lands via reflected light.
4. **B** — Flash/SSD has no moving parts, so it is faster, silent, lower power and more durable.
5. **B** — RAM is volatile, read/write, and holds running programs and data.
6. **C** — ROM holds the bootstrap/BIOS firmware (non-volatile, normally read-only).
7. **C** — Virtual memory uses secondary storage as extra RAM when physical RAM is full.
8. **B** — Tape is serial access — cheap, high-capacity, ideal for archive/backup but poor for random access.

**Section B (12 marks)**

9. **[2]** — 1 mark per difference (max 2): RAM is volatile / ROM is non-volatile; RAM can be read and written / ROM is normally read-only; RAM holds running programs and data / ROM holds bootstrap/BIOS firmware.

10. **[2]** — 1 mark: non-volatile = it **retains its contents even without power** (data is not lost when the device is switched off). 1 mark: any valid example — HDD, SSD, USB flash drive, SD card, CD/DVD/Blu-ray, magnetic tape.

11. **[3]** — 1 mark recommendation: **flash / SSD (e.g. external SSD or USB flash drive)**. Up to 2 marks for justification linked to the requirements: portable (no moving parts, small/light); **fast transfer** speeds suit large video files; **durable** (shock-resistant, no moving parts to damage when carried); easily **rewritable**. (Reject answers that only list generic pros without linking to the scenario; cap at recommendation mark.)

12. **[2]** — 1 mark: when RAM is full, the OS moves less-used **pages** out to secondary storage (swap/paging file) and loads them back when needed, so secondary storage acts as extra RAM. 1 mark drawback: secondary storage is **much slower than RAM**, so heavy paging slows the system / can cause disk thrashing.

13. **[1]** — When excessive paging occurs (pages constantly swapped in and out of disk), the system spends most of its time moving pages rather than doing useful work, slowing it dramatically.

14. **[2]** — 1 mark per advantage (max 2): access from anywhere with internet; scalable capacity (pay for what you need); automatic off-site backup / disaster recovery; easy file sharing/collaboration; no need to buy/maintain local hardware.

**Section C (5 marks)**

15. **[5]** — Point/level marking, up to 5 for justified recommendations that **compare** the two printers against the requirements. Indicative content: - **Office → laser printer.** Fast printing of high volumes; low cost per page for text; sharp/high-quality text — matches "high volume, quick, cheap, black-and-white text". - **Home → inkjet printer.** Cheaper to buy and better for occasional colour photo printing; good colour/photo quality at low volume — matches "occasionally prints a few colour photos". - Comparison ("whereas"): a laser is more expensive to buy and less ideal for occasional colour photos, so it is overkill for the home user; an inkjet is slower and has higher per-page running costs for high volumes, so it is unsuitable for the office. - Justified link of each characteristic (speed, running cost, colour/photo quality, volume) to the stated requirement. Max 4 if the answer states pros without comparing the two printer types / does not justify against the scenario.

**Total: 8 + 12 + 5 = 25**